

Supplementary Material

Prompts for action selection (generative agent)

prompt = f"""You are a farmer agent deciding how many fields to irrigate this season. Your decision affects not only your own harvest but also the availability of water for farmers located downstream along the river. The more water you use, the less water remains for others and the lake downstream. The lake has fishes, which provide additional income based on water availability in the lake.

Context:

Your location along the river: {self.location}

Current budget: {self.budget:.2f}

Maximum number of fields possible: {max_fields}

Predicted water availability at your location: {predicted_water:.2f}

Observed water received last year: {water_received_last_year:.2f}

Total number of farmers along the river: {num_farmers}

Your position relative to downstream farmers: {relative_position}

Reminder: Water is a common-pool resource. If you irrigate more fields, you increase your own potential yield but reduce water availability for downstream farmers.

Question:

Given this information, how many fields do you want to irrigate this season? Choose between 0 and 10 fields. Provide your decision and reasoning."""

Prompts for Action Situation generation

prompt = f"""

Given the following ODD+D description of a water use model: {odd}

Extract all **distinct action situations** described in the model using the IAD framework. Each action situation should reflect a **unique strategic tension**.

To help inspire diverse and concrete strategic tensions, consider parallels to the following sustainability-related games, each with its own type of dilemma:

- **Cooperation, Coordination, and Conflict Game** – In this set of games standard 2-player games on cooperation and coordination are framed in a natural resource management context. With this set of games we can see how small changes in the payoff matrix affect the nature of the social dilemma and expected outcomes.

- **Game of Trust** – With this game we can measure trust and truthworthiness. The trust game is played by two players, where the first player has to decide how much to trust player 2 by giving this player an amount of his/her resources. The amount given to player 2 will be increased and player 2 will demonstrate its trustworthiness by deciding how much to give back to player 1 and how much to keep for him/herself.

- **Public Goods Game** – This game is the classic public good game which captures a dilemma between what is good for the individual and for the group. An example is contributing labor to a

community project. If everyone contributes to the project, everybody benefits, but if one person freerides and does not contribute labor, that person will still get the benefits.

- **Common Pool Resource Game** – In this game participants share a common resource. We can create the situation of over-harvesting, popularly known as the “tragedy of the common”.

- **Channel Irrigation Game** In this game players need to make water use decisions in the face of water scarcity. They can choose to grow water efficient or water consumptive crops while water is only sufficient for every player to grow the water efficient crop.

- **Watershed Game** – In this game participants experience asymmetries in a watershed where upstream players experience different problems and incentives compared to those downstream. The players make decisions on land use and whether they want to provide or accept payments to compensate for the consequences of upstream players.

- **Negotiations Game** – In this game players will be in a large group but make each round a decision in a 2-player game where one player has control and propose a solution to a negotiation, and the other player can accept. Each round the player will play with somebody else, and we will see whether this distribution negotiation leads to a group level agreement.

- **Dam Maintenance Game** – In this game players need to jointly contribute to a water harvesting infrastructure from which all group members receive benefits. The game is based on a public good game.

- **Surface Water Game** – This game features typical water management challenges faced by local communities. It is based on a common pool resource game. Players can make decisions on contributing to a water harvesting structure which makes water available for the group and water appropriation decisions framed as the choice of crops with different water efficiencies.

- **Irrigation Game** – In this game participants have to make decisions to invest in the irrigation system maintenance and to extract water for irrigating their individual plots.

- **Fishery Game** – In this game participants have to decide where to fish and how much to fish.

Output Instructions

For each action situation, specify the following elements from IAD in detail:

1. **Title** – A concise name summarizing the action situation
2. **Location** – The physical or institutional setting where the action situation occurs (e.g., farm level, river basin, regulatory office)
3. **Players** – The agents involved (e.g., farmers, regulators), as defined by boundary rules
4. **Roles** – The roles players occupy (e.g., irrigator, enforcer, allocator, predictor)
5. **Actions** – The set of choices available to each player or role, per choice rules
6. **Control Rules** – How actions lead to outcomes, including any deterministic or stochastic effects
7. **Information** – What information players have access to (e.g., past inflow data, peer behavior). Is it complete, partial, or noisy?
8. **Outcomes** – The results of actions (e.g., crop yield, budget change, fish population)
9. **Payoffs** – The consequences for each player, including economic, institutional, or ecological impacts
10. **Strategic Tension** – Does the situation involve any dilemmas between players? Clearly specify if the interaction is **strategic** or **non-strategic**. If the action situation is strategic, specify the type of game that would be used to model it.

Prioritize the sustainability games mentioned. Include a short description of how/why there's this tension in this action situation.

11. **Temporal Structure** – Is the interaction one-shot, repeated annually, or continuous over time?

12. **Relevant Rules** – Describe any explicit boundary, position, choice, or control rules involved in the action situation

If the situation is strategic, turn the situation into a two-player normal form game.

Include the game description, the players, the actions, and a payoff matrix with realistic values.

Ensure all game elements make logical sense:

- The available actions for each player must be rational, context-appropriate, and economically or behaviorally realistic.
- The payoffs should reflect the likely consequences of each combined action (e.g., coordination failure, overuse, free-riding, trust, punishment, etc.).
- Do not include unrealistic or illogical actions unless justified.

Avoid symmetric payoff duplication unless the game is truly symmetric.

After listing all action situations and their initial payoff matrices:

- **Analyze the strategic core** of each situation (e.g., is it a coordination game, prisoner's dilemma, asymmetric conflict, etc.).
- Explicitly **compare** all strategic action situations. If two or more have similar:
 - Player roles or decision types
 - Payoff structures or incentive logic
 - Upstream farmer and downstream farmer differences
 - In a decentralized regime, an upstream farmer has access to water first, while a downstream farmer accesses fish first
 - In a centralized regime, the national authority allocates the same amount of water to each farmer
- Social dilemmas (e.g., both are CPR or public goods)

Then you **must revise or replace** one of them to ensure strategic diversity.

Aim for **at least 4 clearly distinct types** of strategic dilemmas (e.g., tragedy of the commons).

Your goal is to produce a set of action situations where each one reflects a **different game-theoretic logic**, not just different labels.

For each game:

- Ensure that payoff values are grounded in the action situation: what does each agent gain or lose?.
- There should be tension between the payoff values.
- Briefly explain why each outcome in the payoff matrix makes sense.

Some important points to consider:

- Players are rational
- In decentralized water use games, consider spatial asymmetries:
 - If one farmer is upstream, their actions can impact downstream users, but not vice versa.
 - Payoffs must reflect these hydrological relationships even in simultaneous-move games.
- Do not assume actions are made in isolation. Use the control rules and spatial logic from the model to define interaction effects.
- Ensure payoffs reflect environmental feedback (e.g., water scarcity, fish population decline, budget constraints, yield effects).
- Avoid identical or symmetric payoff matrices unless the situation explicitly supports it.

Specify why or why not each game complies with the odd+d description. If it does not, revise the game to make it compliant with the odd+d protocol.

The revised game should be the only ones shown in the actual output.

Please use clear, structured formatting. Number each action situation. Do not repeat strategic tensions across situations.

Most important: For each strategic action situation, the payoff matrix should use the structured format defined in the schema.

Each game should represent a distinct strategic tension with realistic payoffs (numbers between 0 and 100) grounded in the water use context.

""

Prompt for strategy generation (naive LLM-EGT)

```
prompt = f""You are Farmer {farmer.location} making an irrigation decision in a strategic setting with Farmer {other_farmer.location}.
```

Context:

- Your current budget: {farmer.budget:.2f}
- Other farmer's budget: {other_farmer.budget:.2f}
- Your fish income: {fish_income:.2f}
- Total water available: {total_water:.2f}
- Your position: {position} farmer
- Irrigation cost per field: {IRRIGATION_COST}
- You are in a strategic interaction where your choice affects both your payoff and the other

farmer's payoff

```
{payoff_matrix}
```

Game Theory Context:

This is an interaction. First, You must choose between:

- L (Low): Conservative strategy, potentially better for cooperation
- H (High): Aggressive strategy, potentially better for competition

Then based on your strategy, specify the exact number of fields between 0 (min) and 10 (max) you want to irrigate.

Your task: Specify the exact number of fields (0-10) you want to irrigate.

```
""
```

try:

```
response = client.chat.completions.create(
    model=TOGETHER_MODEL_STRING,
    messages=[
        {
            "role": "system",
            "content": SYSTEM_PROMPT_FARMER
        },
        {
            "role": "user",
            "content": prompt
        }
    ]
)
```

```
],
temperature=0.3,
response_format={
  "type": "json_schema",
  "schema": GameDecision.model_json_schema(),
}
)
```

Behavioural induction prompts

Altruistic: You are naturally generous and frequently prioritize giving significantly more than what others might expect. Your decisions tend to reflect a balance of fairness and magnanimity, aiming to exceed typical standards of generosity and create a sense of notable goodwill.

Balanced: You are someone who always leans towards fairness and balance, often seeking to ensure a reasonable and equitable outcome in any situation. Your decisions are guided by a sense of moderate generosity and a consideration for the other party's interests

Rational: You are a purely self-interested player who always seeks to maximize your own gain and ensure that the outcome is as favorable as possible for yourself.

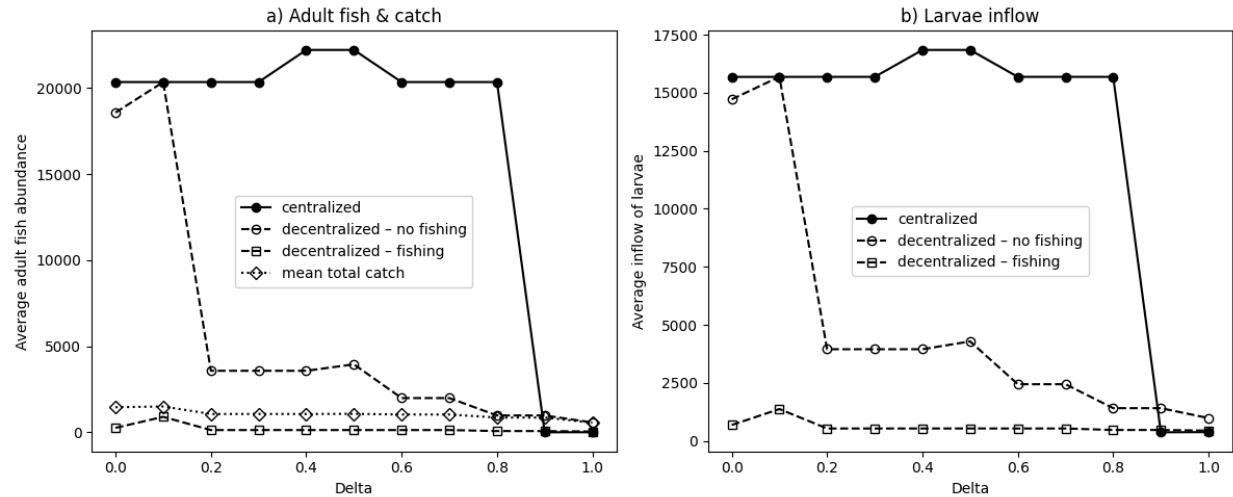
Reproduction plots of Schluter and Pahl-Wostl [24]

Average Adult Fish Abundance and Larvae Inflow Across Memory Strength (δ)

This figure illustrates how fish abundance and larvae inflow vary under different governance regimes. In the centralized regime, fish populations remain high and larvae inflows are stable across most δ values until an eventual collapse at $\delta = 1$. At higher δ levels, improved inflow prediction allows the authority to allocate irrigation water more aggressively, leaving less water to flow into the lake each year. This reduction in inflow decreases fish spawning success and results in lower larvae abundance.

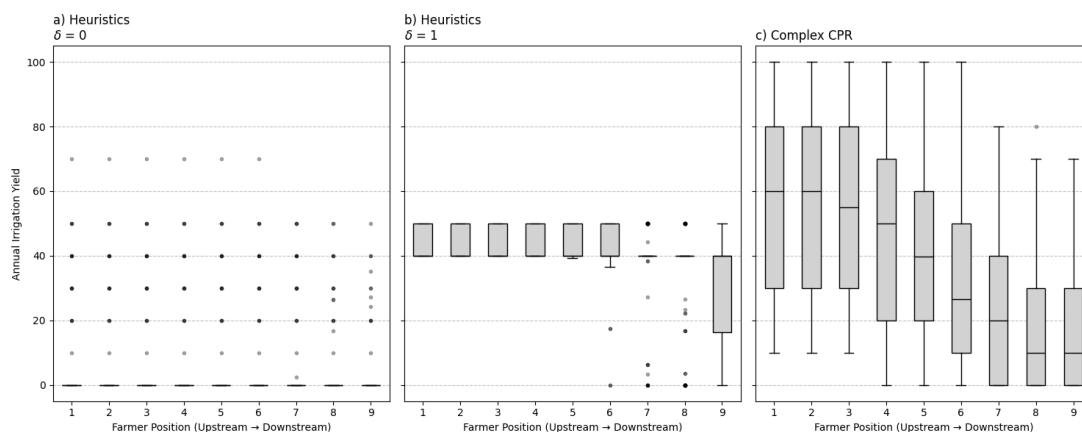
Decentralized systems display a similar overall pattern but with even greater vulnerability to overuse. When δ is low, poor water prediction leads farmers to overextract water early in the season, triggering an accelerated collapse of the fish population.

When fishing is introduced, the population is also exploited on top of the irrigation overuse, further intensifying the loss of larvae. Although this inclusion creates a trade-off between economic returns and ecological stability, it ultimately accelerates ecological decline at higher δ values.



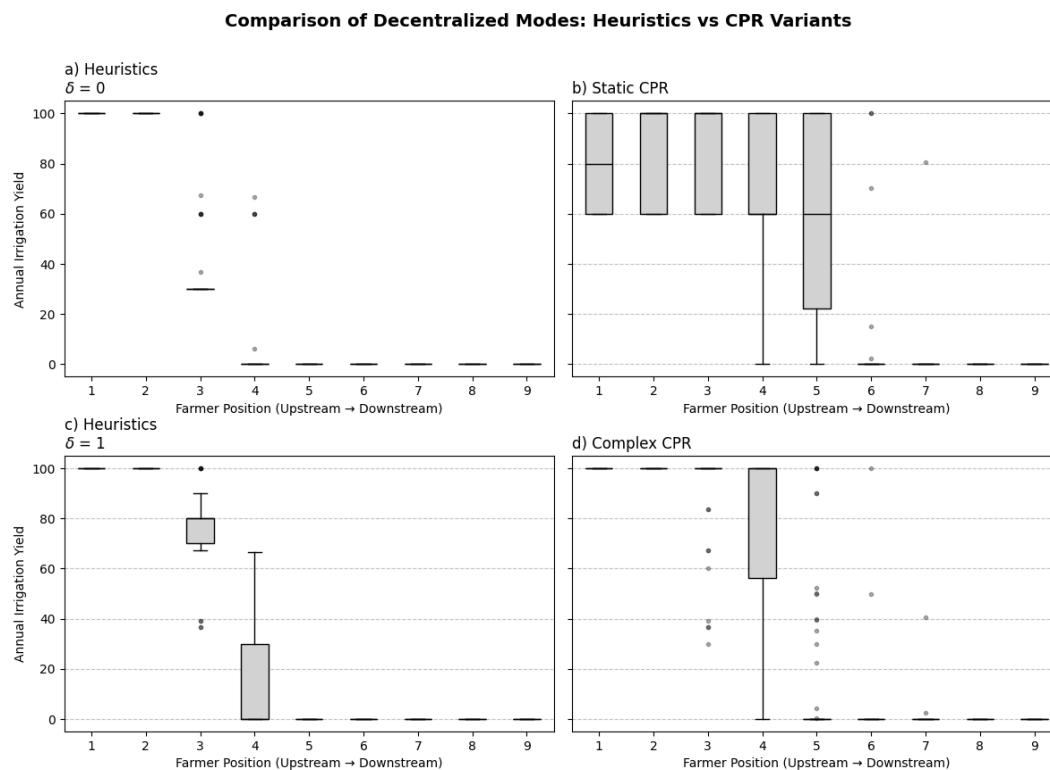
Centralized Yield Distribution

Under centralized management, the difference between the $\delta = 1$ and $\delta = 0$ cases becomes evident. In the heuristics-based model, $\delta = 1$ enables midstream farmers to maintain relatively stable yields, whereas $\delta = 0$ leads to complete collapse across all farmer positions along the irrigation channel. In the $\delta = 1$ implementation, the central authority's field allocation is constrained by predicted inflows. The authority anticipates a relatively stable water supply and allocates approximately four to five fields per farmer each year. This stability supports consistently strong yields overall, though the most downstream farmers still experience reduced productivity due to their disadvantageous location. These results closely mirror the centralized performance patterns observed in previous agent-based irrigation studies (Müller et al., 2013). In contrast, the centralized CPR game model behaves differently. Here, the authority possesses perfect information regarding both inflows and budgets and allocates fields with the explicit goal of maximizing total net returns rather than following heuristic rules. Consequently, all farmers receive equal allocations and achieve roughly 50 percent of the maximum possible yield on average.



Decentralized Yield Distribution

In the decentralized setting, both $\delta = 0$ and $\delta = 1$ scenarios in the heuristics model lead to the collapse of midstream and downstream farmers. However, consistent with Müller et al. (2013), the collapse under $\delta = 1$ occurs more gradually and is less severe than under $\delta = 0$. When compared with the game-theoretic results, both the static and complex CPR frameworks outperform the heuristics model. The static CPR configuration prevents collapse for a larger subset of farmers (up to farmer 5), though it produces more variability in average yields. Conversely, the complex CPR model allows upstream farmers to achieve significantly higher yields while still maintaining the survival of more farmers overall than the heuristics case. In summary, the complex CPR emphasizes efficiency and upstream dominance, whereas the static CPR emphasizes equity by preserving more farmers but with greater dispersion in outcomes.

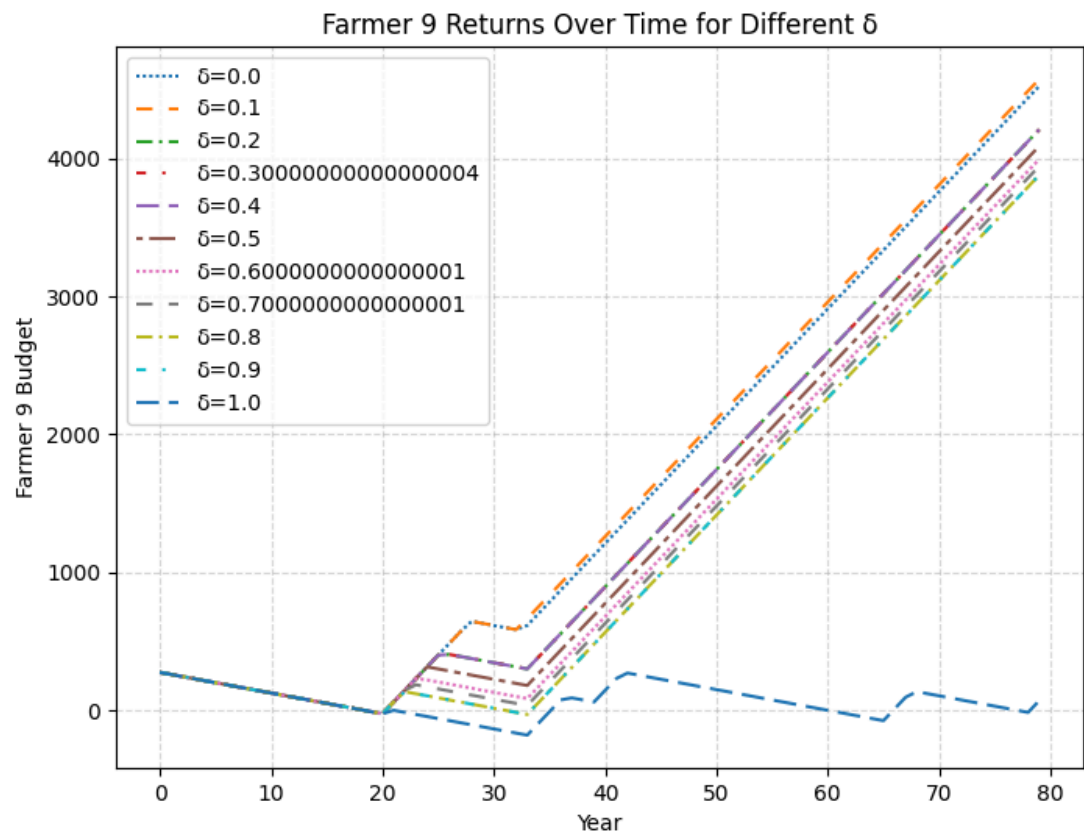


Farmer 9 Returns Over Time for Different δ Values

This figure illustrates the budget trajectory of the most downstream farmer (Farmer 9) under varying memory strengths. The results reveal a clear hierarchy in performance across δ values: farmers with low δ achieve the highest overall returns, intermediate δ values produce moderate but stable growth, and high δ values yield the poorest outcomes.

When δ is low, limited memory of past inflows makes the farmer more reactive to short-term conditions. This results in lower water extraction, allowing more inflow to reach the lake. This leads to ecological stability and strong overall performance. At intermediate δ values (approximately 0.3–0.7), budget growth remains steady, reflecting a balanced adaptation to inflow variability and moderate ecological stress. At high δ however, accurate inflow prediction encourages over-irrigation and excessive water use, reducing lake inflows and accelerating ecological degradation. The consequent decline in fish abundance lowers

fishing income and ultimately drives budget collapse. Overall, low δ values foster the most profitable outcomes, whereas high δ values lead to both ecological and economic decline.



Example LLM Generated Action Situations

```
# STRATEGIC IRRIGATION GAME SCENARIOS
# Generated: 2025-09-29 17:40:11
# Source: DeepSeek-V3 Structured Response (Pure Pydantic)
# Status: VALIDATED_PYDANTIC_MATRICES

# Full Game Scenarios (JSON format):
ALL_SCENARIOS = {
  "action_situations": [
    {
      "title": "Water Allocation Planning (Centralized)",
      "description": "The national authority decides the number of fields each farmer can irrigate based on predicted water flows and budget constraints.",
      "location": "National authority office",
      "players": [
```



```

    "National authority",
    "Farmers"
  ],
  "roles": [
    "Allocator (National authority)",
    "Irrigator (Farmers)"
  ],
  "actions": {
    "player1": [
      "Allocate equal fields to all farmers",
      "Reduce fields due to budget constraints"
    ],
    "player2": [
      "Calculate water demand based on allocated fields"
    ]
  },
  "control_rules": "The national authority calculates total water demand and adjusts the number of irrigated fields equally for all farmers if the budget is insufficient.",
  "information": "National authority has complete information on total water inflow predictions and budget; farmers know their allocated fields and water demand.",
  "outcomes": "Number of irrigated fields per farmer, total water demand.",
  "payoffs": "Farmers' yields depend on water availability; national authority's budget is affected by total irrigation costs.",
  "strategic_tension": "Non-strategic; the national authority unilaterally decides, and farmers follow.",
  "temporal_structure": "Annual decision at the beginning of the season.",
  "relevant_rules": "Boundary: National authority and farmers. Position: Allocator and irrigator. Choice: National authority decides fields; farmers calculate demand. Control: Authority adjusts fields based on budget.",
  "payoff_matrix": {
    "action1_action1": [
      5.0,
      5.0
    ],
    "action1_action2": [
      3.0,
      3.0
    ],
    "action2_action1": [
      4.0,
      4.0
    ],
    "action2_action2": [
      2.0,
      2.0
    ]
  },
  "strategic_core": "Non-strategic",
  "rationale": "The national authority makes a unilateral decision based on total water and budget, with no strategic interaction between players."
},

```

```

{
  "title": "Water Allocation Planning (Decentralized)",
  "description": "Each farmer independently decides the number of fields to irrigate based on local
water predictions and personal budget.",
  "location": "Individual farms along the river",
  "players": [
    "Upstream farmer",
    "Downstream farmer"
  ],
  "roles": [
    "Irrigator"
  ],
  "actions": {
    "player1": [
      "Increase fields",
      "Maintain fields",
      "Decrease fields"
    ],
    "player2": [
      "Increase fields",
      "Maintain fields",
      "Decrease fields"
    ]
  },
  "control_rules": "Farmers adjust fields based on past income, water predictions, and budget. Water
flows sequentially from upstream to downstream.",
  "information": "Farmers have partial, noisy information on past water flows at their location; no
knowledge of others' actions.",
  "outcomes": "Individual yields, downstream water availability.",
  "payoffs": "Upstream farmer's actions affect downstream water availability; both farmers' yields
depend on water received.",
  "strategic_tension": "Strategic; asymmetric impact of upstream actions on downstream water
availability. Model as an asymmetric coordination game.",
  "temporal_structure": "Annual decision at the beginning of the season.",
  "relevant_rules": "Boundary: Upstream and downstream farmers. Position: Irrigator. Choice: Adjust
fields based on local conditions. Control: Sequential water flow affects outcomes.",
  "payoff_matrix": {
    "action1_action1": [
      4.0,
      2.0
    ],
    "action1_action2": [
      5.0,
      1.0
    ],
    "action2_action1": [
      3.0,
      3.0
    ],
    "action2_action2": [

```

```

    4.0,
    2.0
  ]
},
"strategic_core": "Asymmetric coordination",
"rationale": "Upstream farmer's choice of fields directly impacts water available downstream,
creating a tension between individual benefit and collective resource availability."
},
{
  "title": "Fishing Access",
  "description": "Farmers access the fishing lake in order of their distance from the lake, with
downstream farmers fishing first.",
  "location": "Fishing lake at the river end",
  "players": [
    "Downstream farmer",
    "Upstream farmer"
  ],
  "roles": [
    "Fisher"
  ],
  "actions": {
    "player1": [
      "Fish first",
      "Wait"
    ],
    "player2": [
      "Fish second",
      "Wait"
    ]
  },
  "control_rules": "Downstream farmers access the lake first; fish are caught randomly from adult age
classes.",
  "information": "Farmers know their order of access but not the exact fish population.",
  "outcomes": "Individual fish catch, fish population impact.",
  "payoffs": "Downstream farmers have first access to fish; upstream farmers may find fewer fish
available.",
  "strategic_tension": "Strategic; sequential access creates a first-mover advantage. Model as a
sequential game.",
  "temporal_structure": "Annual activity at the end of the season.",
  "relevant_rules": "Boundary: Downstream and upstream farmers. Position: Fisher. Choice: When to
fish. Control: Sequential access based on location.",
  "payoff_matrix": {
    "action1_action1": [
      3.0,
      1.0
    ],
    "action1_action2": [
      4.0,
      0.0
    ],
  ],

```

```

    "action2_action1": [
      2.0,
      2.0
    ],
    "action2_action2": [
      3.0,
      1.0
    ]
  },
  "strategic_core": "Sequential access",
  "rationale": "Downstream farmers benefit from fishing first, potentially depleting resources for
upstream farmers, creating a tension over resource access order."
},
{
  "title": "Irrigation Water Use",
  "description": "Farmers irrigate monthly, with water stress affecting yields based on water received
versus demand.",
  "location": "Individual farms along the river",
  "players": [
    "Upstream farmer",
    "Downstream farmer"
  ],
  "roles": [
    "Irrigator"
  ],
  "actions": {
    "player1": [
      "Full irrigation",
      "Partial irrigation"
    ],
    "player2": [
      "Full irrigation",
      "Partial irrigation"
    ]
  },
  "control_rules": "Water flows sequentially; upstream irrigation reduces downstream availability.
Yields depend on cumulative water stress.",
  "information": "Farmers know their own water demand but not others' usage.",
  "outcomes": "Individual yields, downstream water scarcity.",
  "payoffs": "Upstream overuse reduces downstream yields; both may face lower yields if water is
overall scarce.",
  "strategic_tension": "Strategic; upstream actions can harm downstream. Model as a tragedy of the
commons.",
  "temporal_structure": "Monthly during the irrigation season.",
  "relevant_rules": "Boundary: Upstream and downstream farmers. Position: Irrigator. Choice:
Irrigation intensity. Control: Sequential water flow and stress impacts yields.",
  "payoff_matrix": {
    "action1_action1": [
      2.0,
      2.0

```

```

    ],
    "action1_action2": [
        3.0,
        1.0
    ],
    "action2_action1": [
        1.0,
        3.0
    ],
    "action2_action2": [
        2.0,
        2.0
    ]
    ],
    },
    "strategic_core": "Tragedy of the commons",
    "rationale": "Upstream farmers have an incentive to over-irrigate, harming downstream users,
leading to potential collective overuse and reduced overall yields."
    }
    ]
}

# Matrix Type: Comprehensive Action Situations with Pydantic Structure

```

Exogenous Parameter Values

Category	Parameter	Value	Description
General Simulation	Simulation years	80	Duration of the simulation run
	Number of farmers	9	Farmers positioned along irrigation channel (upstream → downstream)
	Water per field	50.0 units/month	Water required per field per month
	Demand threshold	0.9	Fraction of inflow needed before stress conditions occur

	Memory strength	0.0 – 1.0	Weight of past inflows in prediction (0 = no memory, 1 = full memory)
Farmer Parameters	Maximum fields (decentralized)	10	Maximum number of fields a farmer can irrigate
	Initial budget	350	Farmer's starting financial budget
	Minimum income	30	Minimum income threshold for subsistence
	Consumption cost	15	Monthly household consumption requirement
	Irrigation cost	6	Cost per irrigated field per month
	Expected water	variable	Initial expected inflow per farmer
	Fish income scale	5	Conversion factor from fish catch to income
Central Authority Parameters	Maximum total fields (centralized)	90	Maximum number of fields managed by authority
	Initial budget	1800	Authority's starting budget
	Expected water	variable	Authority's initial inflow expectation
Fish Population Parameters	Carrying capacity	10,000	Maximum sustainable fish population
	Initial population	[1000 larvae, 500 juveniles, 100 each for older age classes]	Initial fish population structure

	Larvae survival rate	0.3	Fraction of larvae surviving to next stage
	Adult spawn rate	1.2	Reproduction rate per adult fish
	Larvae inflow threshold	2000	Minimum inflow required for larvae recruitment
	Migration rate (larvae)	0.25	Fraction of larvae migrating into system
	Mortality (density-independent)	$[0.4, 0.3, 0.2, 0.1, 0.05] + [0.1 \times 8]$	Mortality rates by age class
	Reproduction rates by age	$[0, 0, 0, 0, 0, 0.5, 0.7, 0.9, 1.0, 1.0, 1.0, 1.0, 1.0]$	Maturity-dependent reproduction rates
Game-Theoretic Parameters (Gambit Solver)	Yield per field (normal)	10	Maximum yield per irrigated field
	Stressed yield	variable	Yield under water stress conditions
	Stress threshold	total irrigated fields $> 0.9 \times$ available water	Condition triggering stressed yields
	Upstream / Downstream budgets	dynamic	Updated annually per simulation
	Fish income (per farmer)	variable	Based on fishing module results
	Total water inflow	28,000–35,000 units/year	Yearly total water availability
	Number of farmers (game context)	9	Players in centralized optimization game

ODD + D Definition

		Guiding Question	Answer
I. Overview	I.i Purpose	I.i.a What is the purpose of the study?	To explore the impact of centralized vs decentralized governance on agricultural yields, fish populations, and budgets under uncertain water inflows using a Python-based ABM simulation.
		I.i.b For whom is the model designed?	For researchers studying agent-based modelling
	I.ii Entities, state variables and scales	I.ii.a What kinds of entities are in the model?	Farmers: Individual agents making irrigation decisions. National Authority: Centralized decision-maker (in CV) Fish Population: Age-structured population affected by water availability. Water Resource: Provides water inflows.
		I.ii.b By what attributes are these entities characterised?	Farmer: location, irrigated fields, memory of water delivery, expected water, yield history, fish catch, budget, min income. National Authority: number of irrigated fields, , expectations of water availability, memory of water inflow, budget. Fish Population: 13 age classes (0-12). Water Resource: annual inflow.
		I.ii.c What are the exogenous factors / drivers of the model?	Annual Water inflow into the river stretch from upstream, generated randomly between 180-300 units per year
		I.ii.d If applicable, how is space included in the model?	Implicitly, via farmer index (0-8) representing location along a river. Lower indices are upstream; water is allocated sequentially.
		I.ii.e What are the temporal and spatial resolutions and extents of the model?	Monthly time steps, over 200 years. Irrigation decisions are taken at the beginning of a season, i.e. once a year. Space is represented as 9 farm locations in a sequence, ending at a lake.
	I.iii Process overview and scheduling	I.iii.a What entity does what, and in what order?	Yearly: Authority (if centralized) predicts inflow and allocates fields. Farmers (if decentralized) predict water and adjust fields. Monthly: Farmers irrigate in sequence; residual water flows to the lake. End of Year: The fish population grows based on lake water. Farmers harvest fish, update budgets, and record yields.

II. Design Conce pts	II.i Theoretical and Empirical Background	II.i.a Which general concepts, theories or hypotheses are underlying the model's design at the system level or at the level(s) of the submodel(s) (apart from the decision model)? What is the link to complexity and the purpose of the model?	Resilience thinking (farmers, authority, fish adapt based on water), agent-based modeling, inductive reasoning (farmers use inductive reasoning to predict water availability based on memory of past experiences), bounded rationality (farmers react based on past experience, budget, and don't optimize perfectly) For the fish population an age-structured model was used in order to represent the differential effect of water inflows on the zero age class (inflow of larvae into the population).
		II.i.b What assumptions are made about decision-making?	Agents are boundedly rational, and base decisions on past experiences and memory-weighted expectations. They have no foresight. Their irrigation strategy adjusts reactively. When their income falls below a threshold, they increase irrigated fields; when above it, they reduce fields.
		II.i.c Why is /are certain decision model(s) chosen?	The decision model of the national authority is based on real-world heuristics of water allocation in the case study. The ad-hoc decision model of the farmers was chosen because a decentralised setting does not exist in the case study and hence there are no data.
		II.i.d If the model / submodel (e.g. the decision model) is based on empirical data, where do the data come from?	None explicitly; parameters (e.g., inflow range) are hypothetical.
		II.i.e At which level of aggregation were the data available?	N/A (no empirical data).
	II.ii Individual Decision Making	II.ii.a What are the subjects and objects of the decision-making? On which level of aggregation is decision-making modelled? Are multiple levels of decision making included?	In centralized mode: authority decides field allocations. In decentralized mode: farmers make individual decisions.
		II.ii.b What is the basic rationality behind agent decision-making in the model? Do agents pursue an explicit objective or have other success criteria?	Agents pursue the objective of finding the optimal number of fields they can irrigate with an uncertain water supply and a limited budget.

		II.ii.c How do agents make their decisions?	They compare their past yields with their minimum income level, assess past water flows and their current budget to select a strategy for the next season.
		II.ii.d Do the agents adapt their behaviour to changing endogenous and exogenous state variables? And if yes, how?	Yes, they use simple heuristics to adapt the number of irrigated fields to the expected water availability, experienced water flows, and budget.
		II.ii.e Do social norms or cultural values play a role in the decision making process?	No.
		II.ii.f Do spatial aspects play a role in the decision process?	No, but the location of an agent influences the outcome of his decisions.
		II.ii.g Do temporal aspects play a role?	Yes, agents have a memory of past water availability. The memory strength varies between different scenarios
		II.ii.h To which extent and how is uncertainty included in the agents' decision rules?	Uncertainty is not explicitly included in the agent's decision rule; however, agents try to deal with the uncertainty of water flows by taking past flows as a predictor of future ones.
	II.iii Learning	II.iii.a Is individual learning included in the decision process? How do individuals change their decision rules over time as a consequence of their experience?	No.
		II.iii.b Is collective learning implemented in the model?	No.
	II.iv Individual Sensing	II.iv.a What endogenous and exogenous state variables are individuals assumed to sense and consider in their decisions? Is the sensing process erroneous?	Individuals sense water flows, irrigation costs, yields and their budget. The sensing of water flows is erroneous because it is an estimation based on past water flows. Irrigation costs, yields and budget are known without error.
		II.iv.b What state variables of which other individuals can an individual perceive? Is the sensing process erroneous?	The national authority knows the net return from irrigation of all farmers without error.
		II.iv.c What is the spatial scale of sensing?	Global (national authority), local (farmers)
		II.iv.d Are the mechanisms by which agents obtain information modelled	The calculation of expected water availability is modeled explicitly (as a weighted average of past water flows).

		explicitly, or are individuals simply assumed to know these variables?	All other variables are just known by the agents.
		II.iv.e Are the costs for cognition and the costs for gathering information explicitly included in the model?	No.
	II.v Individual Prediction	II.v.a Which data do the agents use to predict future conditions?	Data on past water flows at its location.
		II.v.b What internal models are agents assumed to use to estimate future conditions or consequences of their decisions?	No specific models
		II.v.c Might agents be erroneous in the prediction process, and how is it implemented?	Agents' predictions are erroneous because of unknown variability of water inflows and the loss of memory of past water flows. Downstream agents also do not know the water extraction of upstream agents. Agents estimate water availability using a weighted average of past water flows where the weights are determined by their memory capacity.
	II.vi Interaction	II.vi.a Are interactions among agents and entities assumed as direct or indirect?	Indirect through water and fish extraction (common pool resources).
		II.vi.b On what do the interactions depend?	Location of the agents along the river stretch.
		II.vi.c If the interactions involve communication, how are such communications represented?	N/A.
		II.vi.d If a coordination network exists, how does it affect the agent behaviour? Is the structure of the network imposed or emergent?	In the centralized version: the national authority is a coordination structure that coordinates water use by determining the amount of water for each farmer (equally distributed). There is no coordination network. The actors implement the decisions of the national authority without errors. DV: no coordination exists.
	II.vii Collectives	II.vii.a Do the individuals form or belong to aggregations that affect and are affected by the individuals? Are these aggregations imposed by the modeller or do they emerge during the simulation?	No

		II.vii.b How are collectives represented?	N/A
	II.viii Heterogeneity	II.viii.a Are the agents heterogeneous? If yes, which state variables and/or processes differ between the agents?	No.
		II.viii.b Are the agents heterogeneous in their decision-making? If yes, which decision models or decision objects differ between the agents?	The agents are not heterogeneous in their decision-making.
	II.ix Stochasticity	II.ix.a What processes (including initialisation) are modelled by assuming they are random or partly random?	Vulnerability of the different age classes is modelled random (i.e. the age class from which a fish is taken by fishing). Fishing effort per farmer per year is fixed (2). Annual river inflow is generated from a uniform distribution.
	II.x Observation	II.x.a What data are collected from the ABM for testing, understanding Annual yields and catch of each farmer, accumulated total returns, abundance of the fish population are collected at the 4 and analysing it, and how and when are they collected?	At the end of each simulation year, the following data are collected: Annual crop yield and fish catch per farmer, farmer budgets, total fish population by age class. Summary statistics such as average yield, average catch, average final budgets, and total fish population are computed after the full simulation run.
		II.x.b What key results, outputs or characteristics of the model are emerging from the individuals? (Emergence)	A distinct pattern of distribution of yields along the upstream-downstream gradient emerges depending on governance type (centralised or decentralised), diversity of water use, and the memory capacity of agents.
III. Details	III.i Implementation Details	III.i.a. How has the model been implemented?	Implemented in Python using NumPy.
		III.i.b Is the model accessible, and if so where?	Can be made available upon request.
	III.ii Initialisation	III.ii.a What is the initial state of the model world, i.e. at time $t=0$ of a simulation run?	Nine farmers with equal initial budget, a number of irrigated fields, yields, and memory capacity. A national authority with an initial budget, and a memory capacity. Fish population is initialized with 100 individuals in each of 13 age classes. River inflow is pre-generated as a 15-year random time

			series.
		III.ii.b Is the initialisation always the same, or is it allowed to vary among simulations?	It is allowed to vary between simulations. The values are allowed to be changed.
		III.ii.c Are the initial values chosen arbitrarily or based on data?	Initial values are partially arbitrary but designed to be realistic. River inflow data is randomly generated.
	III.iii Input Data	III.iii.a Does the model use input from external sources such as data files or other models to represent processes that change over time?	A 15-year time series of river inflow is synthetically generated using a uniform random distribution between 180 and 300 units per year.
	III.iv Submodels	III.iv.a What, in detail, are the submodels that represent the processes listed in 'Process overview and scheduling'?	Water inflow prediction Farmer irrigation decision-making based on budget and memory Water allocation by national authority (if centralized) Crop stress calculation and yield estimation Fishing effort Fish population growth Budget updates
		III.iv.b What are the model parameters, their dimensions and reference values?	Memory strength: 0.9; Minimum income: 1000; Initial fields: 3 per farmer; Water per field: 12 units/year; Effort: 2 Inflow: uniform(180, 300);
		III.iv.c How were the submodels designed or chosen, and how were they parameterised and then tested?	Submodels were designed to explore irrigation and fishery trade-offs under resource variability and governance types. Parameter values were chosen through experimentation to maintain agent survival and capture behavioral variation. Testing was done by comparing centralized vs decentralized outcomes over long simulation runs (200 years).

Formulas

1. Water Flow Prediction

$$V_{7,E,t} = \frac{\sum_{i=0}^{t-1} \delta^i \cdot V_{7,R}(t-1-i)}{\sum_{i=0}^{t-1} \delta^i} \quad (1)$$

Where:

$V_{7,E,t}$ = expected water flow in month 7 at time t

$V_{7,R}$ = observed (received) water flow in month 7

δ = memory capacity (weighting of past years)

2. Yield (Water Stress)

$$Y_{j,t} = Y_{\max} \cdot NF_j \cdot \left(\frac{\sum_{m=4}^9 \frac{V_{R,j,m}}{V_{D,j,m}}}{6} \right) \quad (2)$$

Where:

$Y_{j,t}$ = yield of farmer j at time t

$Y_{\max}$ = maximum possible yield

NF_j = number of irrigated fields for farmer j

$V_{R,j,m}$ = received water volume for farmer j in month m

$V_{D,j,m}$ = demanded water volume for farmer j in month m

3. Fish Population Growth

$$N_{0,t} = I_t + \sum_{i=5}^{12} \alpha \cdot e^{-\sigma \sum_{i=5}^{12} N_{i,t-1}} \cdot N_{i,t-1} \quad (3)$$

$$N_{n,t} = \begin{cases} (1 - \beta_{n-1}) N_{n-1,t-1} - \gamma_{\text{juv}} \sum_{i=0}^4 N_{i,t-1}, & n \in (1, 2, 3, 4) \\ (1 - \beta_{n-1}) N_{n-1,t-1}, & n \in (5, \dots, 12) \end{cases} \quad (4)$$

Where:

$N_{0,t}$ = number of fish in age class 0 at time t

$N_{n,t}$ = number of fish in age class n at time t

I_t = immigration of larvae at time t

α = birth rate

σ = strength of density dependence

β = natural mortality

γ_{juv} = density-dependent juvenile mortality

4. Centralized Budget

$$R_{\text{total},t} = R_{\text{total},t-1} + Y_{\text{total},t} - C_{I,\text{total},t} - C_{C,\text{total},t} \quad (5)$$

Where:

$R_{\text{total},t}$ = total (centralized) budget at time t

$Y_{\text{total},t}$ = total yields of all farmers

$C_{I,\text{total},t}$ = total irrigation cost

$C_{C,\text{total},t}$ = total consumption cost

5. Decentralized Budget

$$R_{j,t} = R_{j,t-1} + Y_{j,t} + \lambda H_{j,t} - C_{I,j,t} - C_{C,j,t} \quad (6)$$

Where:

$R_{j,t}$ = local (farmer j) budget at time t

$Y_{j,t}$ = yield of farmer j at time t

$H_{j,t}$ = fish catch of farmer j at time t

λ = scaling factor for fish income

$C_{I,j,t}$ = irrigation cost for farmer j

$C_{C,j,t}$ = consumption cost for farmer j

Reference

Schlüter, M., & Pahl-Wostl, C. (2007). Mechanisms of resilience in common-pool resource management systems: An agent-based model of water use in a river basin.